

## UVM FIFO Project

Create a full UVM environment for the FIFO implemented in the SV project.

### Steps:

- Create a **full environment** for the FIFO design.
- Add **constraints** in the sequence item class.
- Add **covergroups, coverpoints** in the coverage collector class.
- Add **assertions and bind it in the top module.**
- Split the FIFO main sequence into multiple sequences based on your verification plan, for example:
  - o **write\_only\_sequence**
  - o **read\_only\_sequence**
  - o **write\_read\_sequence**
- Note: You must not stick to the work done in your SV project. You are free to add or modify in the assertions, covergroups or constraints to enrich your verification.

### Requirements:

- Verification plan
- **Draw your UVM testbench** showing the UVM structure using **powerpoint, draw.io or MS Visio**
  - o Write a section where you will describe in details how the UVM testbench work, from the top module then driving the interface then monitoring and the analyzing the output
- **Code Coverage report**
- **Functional Coverage report**
- **Sequential Domain Coverage report**
- **Bug report**
- Sections with QuestaSim snippets to for each UVM sequence and how the interface is driven in each sequence using the waveform snippets

### Submission file:

.rar file containing the following:

- PDF file having the requirements
- Testbench and design files
- Do file to run simulation (The project will not be graded if the do file is not working or missing)

Also, provide a table in your PDF file showing the assertions used as follows

| Feature                                         | Assertion                         |
|-------------------------------------------------|-----------------------------------|
| Whenever the FIFO is full, wr_ack is always = 0 | @(posedge clk) (full  -> !wr_ack) |